

NATIONAL TECHNICAL RESEARCH ORGANISATION

SIGNAL ANALYSIS & PARAMETER EXTRACTION DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: 4f3e716eb3fb2212

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : sample_16_qam.iq
File SHA-256 Digest : 4f3e716eb3fb2212eb1f71b041633c8d245877742bd45ead6077dbac50956c2e
Payload File Size : 524,288 bytes (512.0 KB)
Analysis Timestamp : 2026-09-18T18:43:11Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	20.0000 MSps	Baseband Nyquist Rate
Center Frequency (F_c)	+1187.113 kHz	Spectral Centroid Offset
Estimated Baud Rate	3.66 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	14.65 kHz	Half-Power Frequency Span
Occupied Bandwidth (99%)	2978.52 kHz	Cumulative PSD 99% Energy
Estimated SNR	4.27 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	5.73 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	+1250000.00 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.0062	Wiener Entropy Index (0.0 – 1.0)

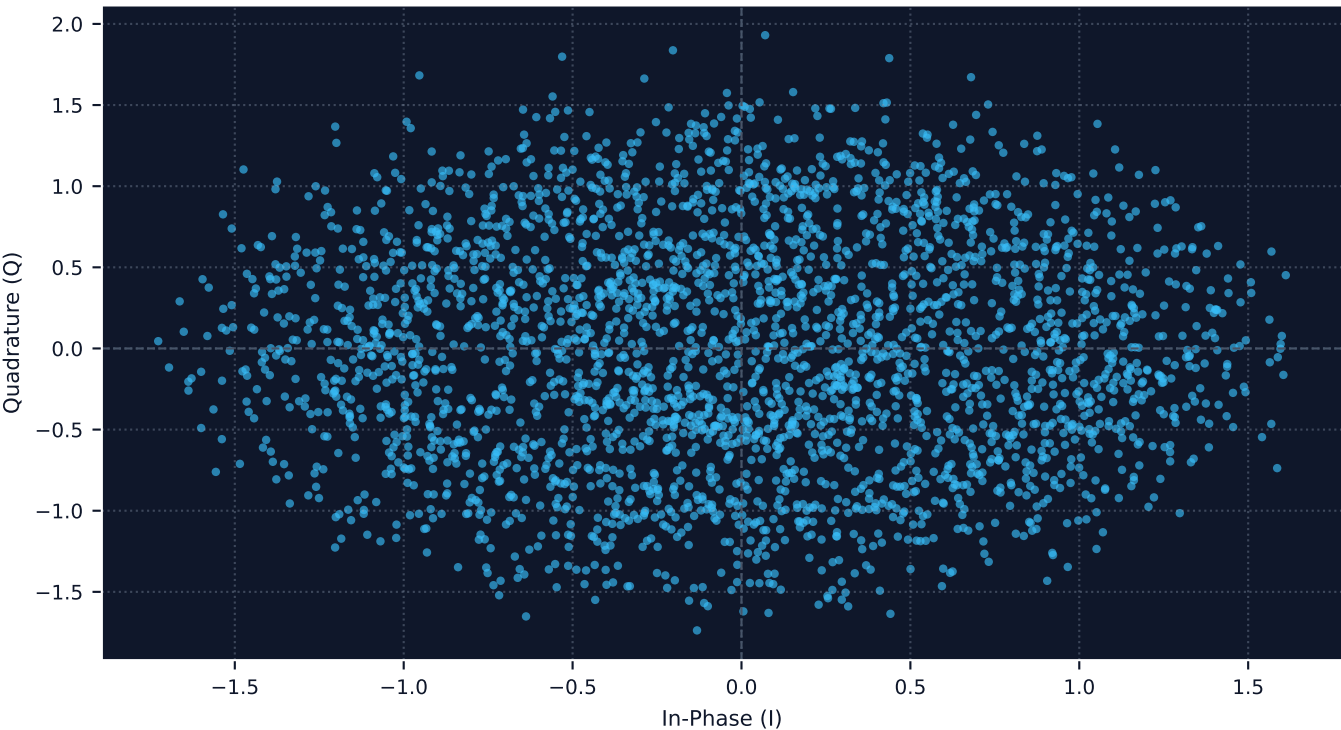
PREDICTED MODULATION : 16-QAM

CLASSIFIER CONFIDENCE : 72.00% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : WARNING: LOW CONFIDENCE CLASSIFICATION

Demodulation Scheme: 16-QAM | Recovered Bits: 262,144 | Estimated EVM: 10.00% | CRC Validated Frames: 0

Demodulated Baseband Constellation Diagram — [16-QAM]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	260	0x55F6	26312	5021 B	CRC MISMATCH	CRC_MISMATCH
#2	554	0x9882	25677	32692 B	CRC MISMATCH	PARTIAL_RECOVER
#3	1050	0x1188	9828	32630 B	CRC MISMATCH	PARTIAL_RECOVER
#4	1226	0x1882	8874	32608 B	CRC MISMATCH	PARTIAL_RECOVER
#5	1650	0x88A6	19955	4488 B	CRC MISMATCH	CRC_MISMATCH

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 FF666666EC8A339DFF66EECC8A319F66 [FRAME 1 PAYLOAD]